

Reranking Improves Factualty on XSum

Factuality-aware reranking for extreme summarization

Bounded study: fine-tuned BART, beam candidates, and post-generation reranking

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What this project is actually doing

A few quick terms make the rest of the talk easier to follow.

- Start with one article and generate several possible summaries instead of trusting only the single top-likelihood output.
- Score those candidates with factuality-oriented signals, then choose the best validation operating point.
- The goal is not a new model family; it is testing whether better selection improves groundedness.

XSum

Single-sentence news summarization benchmark with aggressive compression.

Beam search

A decoding method that keeps several likely next-word paths, giving multiple candidate summaries.

Reranking

Post-generation selection: choose among candidates using extra scores after decoding.

ROUGE

An overlap metric: it rewards wording similarity to the reference summary, not factual correctness by itself.

Why this matters

If reranking works, factual quality can improve without replacing the generator itself.

Why fluent XSum summaries can still be wrong

The goal was better factual support without hiding the overlap trade-off.

- XSum rewards extreme compression, so fluent summaries can still insert unsupported entities or relations.
- ROUGE captures overlap, but not whether a claim is actually grounded in the source article.
- The baseline is the public BART XSum model, so the gain comes from selection rather than a different generator.

Research question

Can reranking trade a little ROUGE for meaningfully stronger factuality?

Baseline

Public BART baseline
beam search only
no factuality fusion

Scope

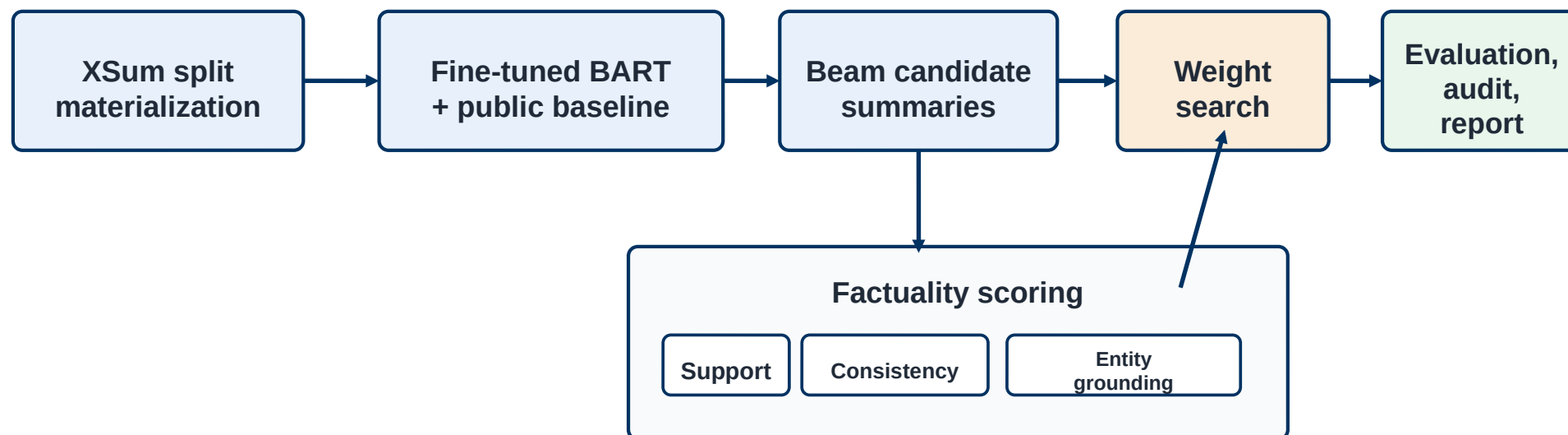
train 128
val 64 / 128
test 128
qualitative 24

Success criterion

Higher factuality composite on the bounded test split, with the ROUGE cost reported rather than hidden.

Generate candidates, score them, choose the operating point

The method is intentionally simple: no new model family, just better candidate selection.



Scoring signals

Support, consistency, and entity grounding are scored for each candidate summary.

Selection rule

Validation tunes the weights once; the chosen setting is then evaluated once on test.

Beam sizes 4, 8, and 16 were explored; the winner was a factuality-heavy beam-16 setting.

RESULTS

Reranking improves factuality, with a modest ROUGE trade-off

The chosen point sits near the high-factuality knee, not the highest-ROUGE corner.

Factuality composite

0.331 → 0.434

+0.103 on bounded test split

ROUGE-Lsum

0.359 → 0.339

-0.020 ROUGE-Lsum trade-off

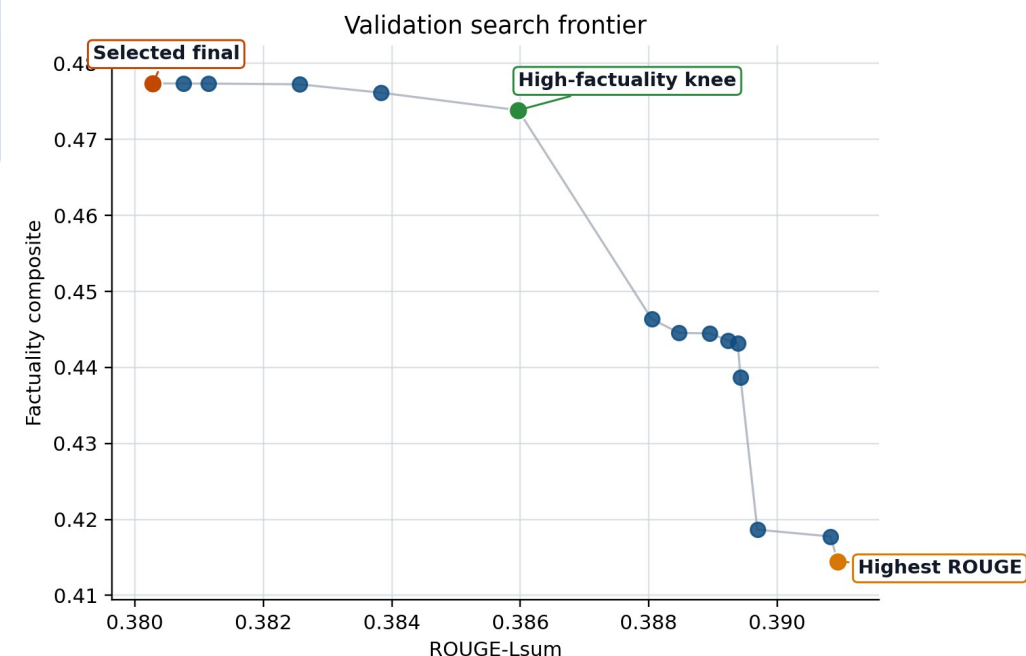
Bootstrap deltas

Factuality CI: [0.082, 0.125]

ROUGE CI: [-0.033, -0.006]

Chosen point

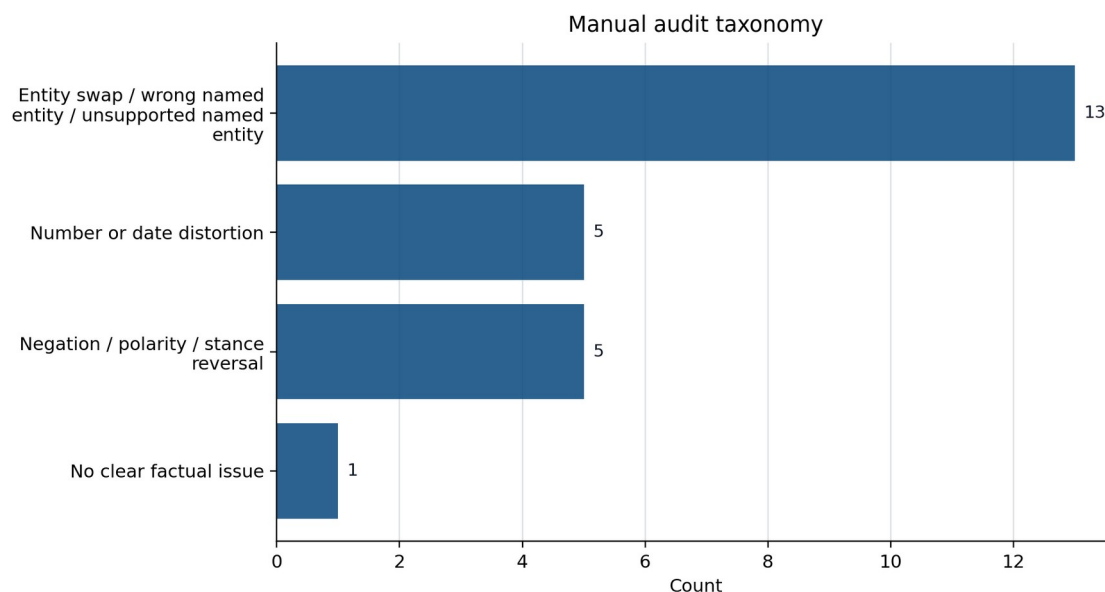
Beam 16 winner; factuality-heavy fusion beat likelihood-heavy settings.



Selected final stays near the high-factuality knee rather than the highest-ROUGE point.

The main remaining failure mode is entity distortion

The reranker helps, but most of the remaining misses are still entity errors.



- 24-example stratified review with Codex / AI-assisted expert adjudication.
- Outcome split is balanced: reranker 8, baseline 8, tie or close 8.
- Entity distortion dominates the remaining errors (13 of 24); number/date and polarity are smaller.

Bounded claim

Directional improvement, not benchmark-scale proof: this supports the approach, not a claim about XSum overall.

Reranking helps, but entity grounding is still the bottleneck

Useful gain now; the strongest next step is better entity preservation.

1 Main result

A lightweight reranker materially improves factuality on this bounded XSum run.

2 Main limitation

Entity-level mistakes still dominate when all beam candidates drift toward the same wrong fact.

3 Next iteration

Broaden candidate diversity and add stronger entity-preservation constraints before scaling the evaluation.

Why this matters

It shows that a small, interpretable post-generation layer can improve factual quality without retraining a whole new architecture.

If I had more time

Larger bounded splits
more diverse candidate sets
stronger entity-aware constraints
external factuality checks

Final message: reranking improves factuality meaningfully, but entity grounding remains the hardest problem.

Questions?